2023 Ph H1 Q13

Section: Particles and Waves

Topic: Wave-Particle Duality

Question Summary:

A student makes three statements about wave-particle duality:

I. The photoelectric effect is evidence supporting the particle model of light.

II. Interference is evidence supporting the wave model of light.

III. Photons of sufficient energy can eject electrons from the surface of metals.

Which of these statements are correct?

Step-by-Step Solution:

• Statement I: ✔ True. The photoelectric effect shows that light behaves as particles (photons), since electrons are ejected only if the photons have enough energy $E = hf$.

• Statement II: ✔ True. Interference patterns (double slit, diffraction gratings, etc.) can only be explained if light behaves as a wave.

• Statement III: ✔ True. This is the very definition of the photoelectric effect: photons above the threshold frequency can eject electrons from a metal surface.

Final Answer:

E — I, II and III

Revision Tips:

- Wave-particle duality means light (and other quantum particles) can show both wave-like and particle-like behaviour depending on the experiment.
- Evidence for particle nature: photoelectric effect, Compton scattering.
- Evidence for wave nature: interference, diffraction, refraction.
- Always link the photoelectric effect to photons and interference to wave superposition.